Ch. 4
Capitalizing on Expected Exchange Rate Movement

FX- foreign currency

-> If you expect foreign currency to appreciate, you can borrow funds in your local currency and buy the foreign currency, you can deposit the funds in a short-term savings account.

-> If you expect foreign currency to depreciate:
  -> borrow currency abroad
  -> buy USD (or your local FX)
  -> deposit in USA bank account
  -> Buy foreign FX when it goes down & repay the loan

Ex. Expectation of N2$ exchange are as follows:

$$S_0 = \$0.50/N2$$

$$E(S_1) = \$0.52/N2$$

1) Is foreign currency going to appreciate or depreciate?

2) Buy N2, deposit in foreign bank account

You have $20 million, interest rates are as follows:

	Lending Rate	Borrowing Rate
USD	6.72% p.a.	7.20%
NZD	6.48%	6.96%

Borrowing rate - rate of a loan
Lending rate - deposit rate

1 month transaction

Borrow USD at 7.2%

1 month later pay back

$20000000 \cdot \left(1 + \frac{7.2\%}{12}\right) =$

$= \$20,120,000$

cost $= \$120000$

Convert $20 million to NZ$

~~$20m~~ : ~~$0.50/NZ~~

NZ$40,000,000$

Deposit for 1 month in N2 at 6.48%

Withdraw

$N2\,40{,}000{,}000 \cdot \left(1 + \frac{6.48\%}{12}\right)$

$N2\,40{,}216{,}000.00$

Convert back to US

$N2\,40{,}216{,}000 \cdot \$0.52/N2$

$= \$20,912,320$

$\pi = \$20,912,320$

$- \$20,120,000$

$792,320$

(Tip: if you did it the other way around, you would get $-\$816,320$:-)

CARRY TRADE

Capitalize on different interest, when currencies are stable

$$S_0 = E(S_1) = \$1.20/€ \quad \$1.80/£$$

	Bid Lending %	Ask Borrowing %
Germany	5.5% p.a.	6%
GB	12%	14%

1) Borrow €1,000,000 in Germany at 6%

2) Convert to GBP & deposit at 12%

3) Bring back for profit

After 1 MONTH

1) Cross-Rate Conversion

$\frac{\$1.20/€}{\$1.80/£} =$

$$= \frac{1.20}{1.80} \text{ £/€}$$

$$= \text{£}0.6667/€$$

Borrow

$$€1{,}000{,}000 \cdot \left(1 + \frac{6\%}{12}\right)$$

$$= €1{,}005{,}000.00$$

0.5% c.p.

Convert & Deposit in British

$$€1{,}000{,}000 \cdot \text{£}0.6667/€$$

$$\text{£}666{,}666.67$$

After a month withdrawal

£666,666.67 · $\left(1+\frac{12\%}{12}\right)$

£666,666.67 · (1.01)

= £673,333.34

Convert back to €

£673,333.34 · £0.67/€

= €1,004,994.96

π = − €1,000,000.00

= €4,994.96.

$1.20/€

€4,994.96 · $1.20/€

= $5,993.95

You expect Australian Dollar to move as follows

$S_0 = \$0.50/AUD$

$E(S_1) = \$0.48/AUD$

If the bank rates are as follows:

	Lending	Borrowing
USD	6.72%	7.20%
AUD	6.48%	6.96%

Explain & calculate how you could profit from this expectation over 1 month.

Use 40 million of local currency